

Co-curricular Courses: CC		Field Project/ RP/CC/Internship/ Apprenticeship/ Community Engagement and Services: FP / CEP	
Type: DSC1-3 (Major) B.Sc(ECS)-II (Semester III) Course Title: Data Structure using C++ (Paper Code:)			
Credits: Theory – (2)		Practical's – (1)	
Total Lectures: 30 Hrs.		Contact Hrs. (L) : 2	
University Evaluation: 30 Marks		Internal Evaluation: 20 Marks	
Course Outcomes:			
1.	Understand basic data structures such as arrays, linked lists, stacks, and queues.		
2.	To be familiar with fundamental data structures and with how these data structures can best be implemented, become accustomed to the description of algorithms in both functional and procedural styles.		
3.	To know the complexity of basic operations like insert, delete, and search on these data structures.		
4.	Ability to choose a data structure to model any data used in computer applications suitably.		
5.	To solve problems involving graphs and trees.		
Unit	Content	Lect.	
I	An Introduction to Data Structures: Introduction, Definition and Types of Data Structure. Abstract Data Type (ADT). Array: Single and two-dimensional arrays and operations Stack: Introduction to Stack, Operations of stack- Create, isempty, isfull, push, pop, display. Implementation of stack using an array (Static Implementation), Applications of Stack-Conversion of infix expression to postfix expression, Conversion of infix expression to prefix expression, Matching parenthesis in an expression (Checking expression is valid or invalid), Evaluation of postfix expression, Stack in recursion, Implementation of all applications of stack. Queue: Introduction to Queue, Operations of the queue- Create, isempty, isfull, insert, remove, display, Types of Queue- Linear Queue, Circular Queue, Deque (Double Ended Queue), Priority queue. Implementation of all types of queue using the array (Static Implementation), Difference between stack and queue, Applications of Queue. Searching and Sorting: Bubble sort, Selection sort, Merge sort, Radix sort, Quick sort, Linear search, Binary search, Complexity of these algorithms	15	
II	Linked Lists: Introduction to Linked Lists, Difference between Array and Linked list. Types of linked list- 1) Linear linked list- Singly (Single) linear linked list and Doubly (Double) linear linked list. 2) Circular linked list- Singly (Single) circular linked list and Doubly (Double)	15	

	circular linked list, Implementation of all types of linked list with operations- creation, insertion, deletion, traversing, searching, display, count, reverse, Implementation of stack using linked list (Dynamic stack), Implementation of queue using linked list (Dynamic queue), Applications of linked list. Tree: Introduction to Tree, Introduction to Binary Trees, Types of Binary tree- Strictly Binary tree, Complete Binary tree, Extended (2-Tree) Binary tree, Binary expression tree, Binary Search tree, Heap Tree- Min heap tree, Max heap tree, AVL tree, Representation of Binary tree using- Array, Linked list, Operations of Binary search tree-Creating and inserting node, Searching node, Counting total nodes, Counting and displaying leaf nodes, Tree Traversal methods Preorder, Inorder, Postorder, Deletion of Nodes, Implementation of binary search tree, Applications of tree.	
Reference Books:		
1.	Tanenbaum: Data structures using C and C++	
3.	Data Structures and Algorithm Analysis in C++ by Weiss, Weiss Mark Allen	
4.	Data Structures Through C++ by Yashavant Kanetkar	